

EuRepoC CYBER INCIDENTS

Exploratory Data Analysis Report

European Repository of Cyber Incidents • Global, Receiver, Attribution & Dyadic Datasets

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OVERVIEW

3,414 TOTAL INCIDENTS	59.2% ATTRIBUTED	83 CLEAN COLUMNS	2000–24 TIME SPAN
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This section evaluates the data integrity and processing framework applied to the 3,414 documented cyber events from 2000 to 2024. It examines the comparative landscape of raw versus refined datasets, highlights the significant mitigation of data sparsity across critical cells, and identifies the variables undergoing the most rigorous enhancement to ensure the structural reliability of the [EuRepoC Global Dataset](#) before proceeding with advanced analytical modeling.

◆ KEY INSIGHT

To avoid an artificial contraction of the historical threat landscape, the [EuRepoC Global Dataset](#) methodology prioritizes rigorous data enrichment over the deletion of sparse records. By maintaining the integrity of documented incident volumes while systematically remediating data gaps across 83 refined variables, this framework captures the full scale of digital aggression and enhances structural reliability for advanced geopolitical and tactical modeling.

EXPLORATORY DATA ANALYSIS

01 Incidents by initiator Country

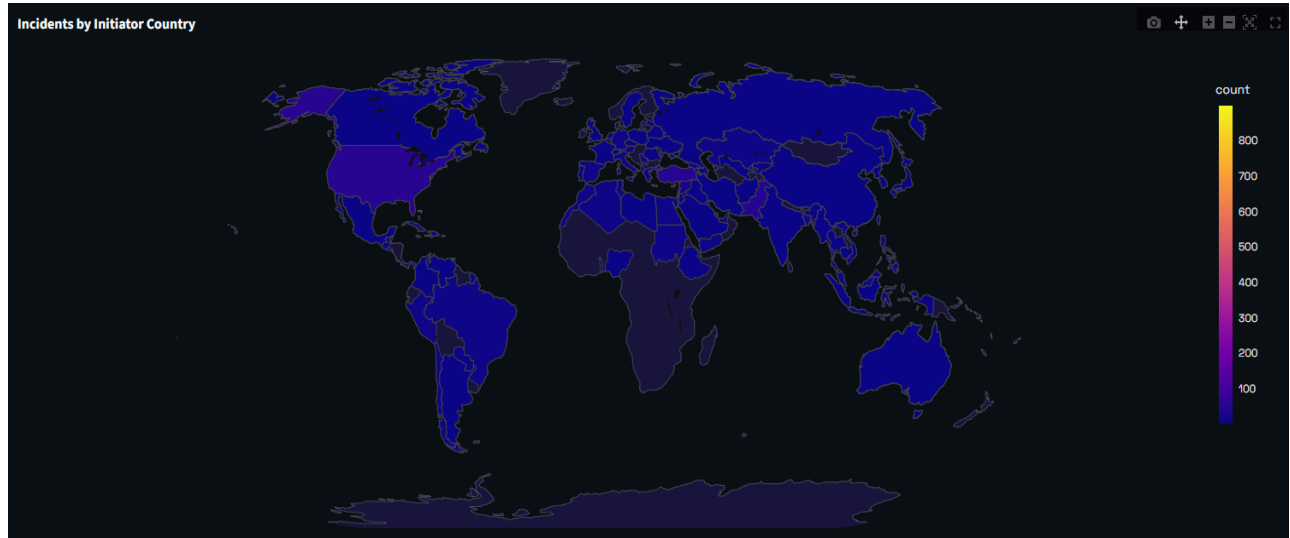


Figure 1. Incidents by initiator Country.

The map of "Incidents by Initiator Country" depicts a highly concentrated global threat landscape, where the vast majority of nations show negligible activity, while a handful of countries stand out as high-frequency initiators.

◆ KEY INSIGHT

This geographic polarization confirms that cyber-offensive capability is not a commodity, but a specialized strategic resource restricted to a limited tier of nations with the requisite infrastructure and political imperative to project power digitally.

02 Incidents by Receiver Countries

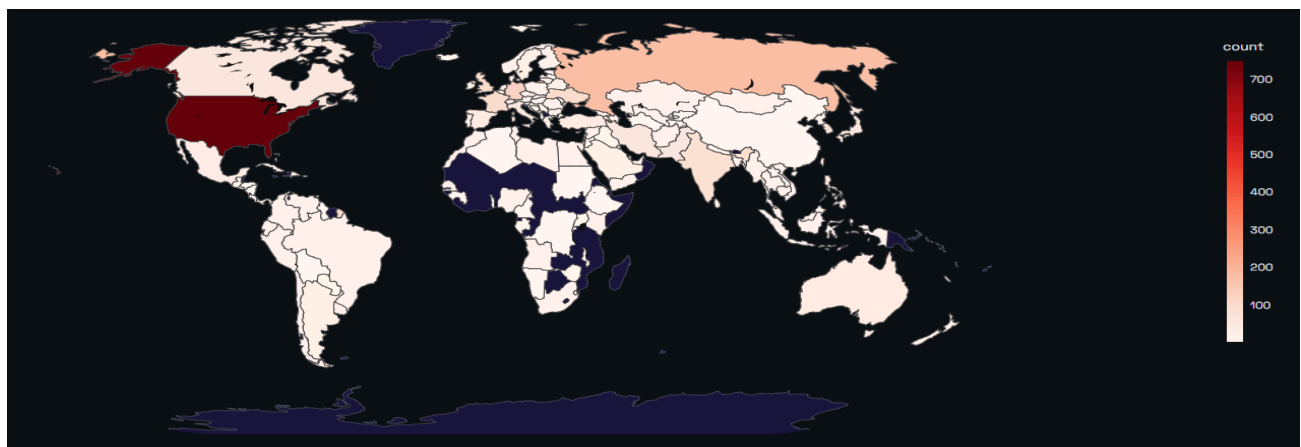


Figure 2. Incidents by Receiver countries.

The map of "Incidents by Receiver Countries" illustrates a stark concentration of risk where a small number of Western and major geopolitical powers absorb the overwhelming majority of global cyber-targeting, while the rest of the world remains relatively untouched.

◆ KEY INSIGHT

This intense geographic clustering suggests that cyber-targeting is not indiscriminate, but rather follows the fault lines of strategic importance, with the most economically and politically significant nations functioning as the primary "shock absorbers" for global cyber aggression.

03 Functional vs Intelligence Impact



The matrix is a heatmap that illustrates the relationship between the functional impact of cyber incidents (measured by system disruption duration, such as days or months) and their intelligence impact (measured by the severity of data breaches or corruption).

◆ KEY INSIGHT

The most striking takeaway is the massive attribution and reporting gap: an overwhelming majority of incidents (**1,433**) are categorized as "**Unknown**" for both functional and intelligence impacts.If we look only at the *known* data, the highest concentration of incidents involves "**No system interference/disruption**" paired with either "**No data breach**" (376 incidents) or "**Minor data breach**" (373 incidents). This suggests that while many attacks occur, the majority of the clearly categorized ones result in relatively low-level overt damage or are successfully mitigated before causing major, long-lasting disruptions.

04 Zero Day Usage

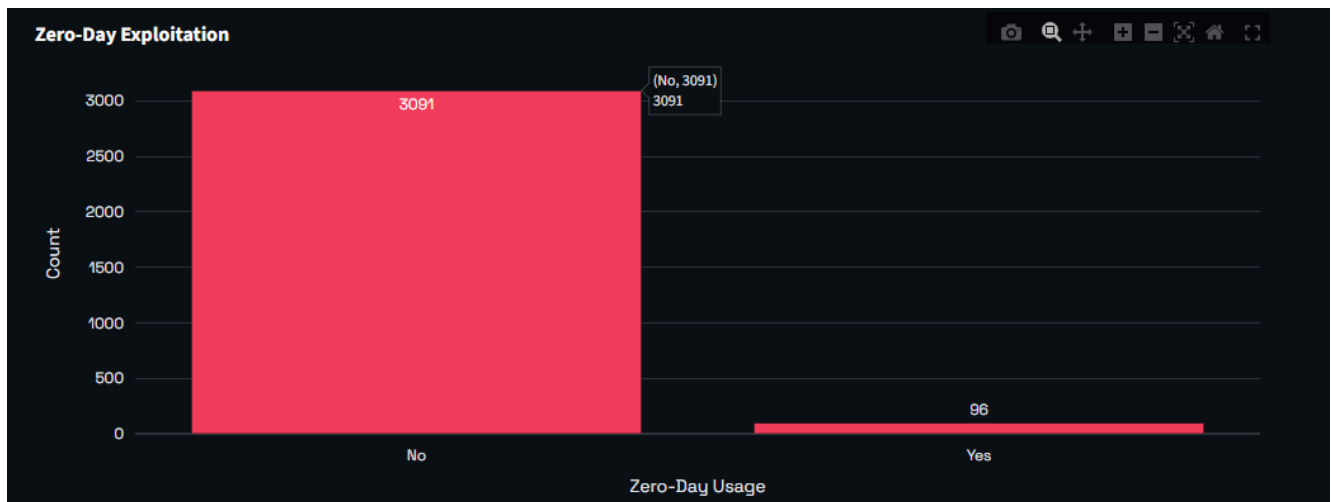


Figure 4. Zero Day Exploitation.

The visualization of "Zero Day Exploitation" underscores a lopsided technical landscape, where the overwhelming majority of cyber activity (3,091 incidents) leverages known vulnerabilities rather than zero-day exploits, which remain a rare tactical choice appearing in only 96 recorded cases.

◆ KEY INSIGHT

The pervasive use of non-zero-day techniques demonstrates that adversaries mainly capitalize on existing security flaws, system misconfigurations, or psychological manipulation instead of utilizing original exploits. Since such novel vulnerabilities are arduous to identify, costly to procure, and depreciate instantly upon discovery, they remain infrequent tactical assets—typically withheld by sophisticated actors for elite objectives that resist conventional penetration methods.

05 Receiver Category — Who Gets Targeted?



Figure 5. Top receiver categories, unpacked receiver dataset.

The analysis indicates a pronounced left-skewed distribution regarding the severity of cyber activity. Within the **Weighted Intensity Distribution**, a significant majority of documented cases fall within the 1 to 5 range on a decile scale. This trend is further validated by the **Impact Indicator Labels**, where entries designated as "Low" impact (1,436 instances) vastly exceed the frequency of "Minor" (395) or "Medium" (70) classifications.

◆ KEY INSIGHT

The visualization reveals a heavily left-skewed landscape of severity, where the vast majority of documented operations result in low-level functional disruptions or minor data compromises. This pattern underscores that the digital threat environment is characterized primarily by persistent, moderate-impact activities—like routine intelligence gathering or localized service interference—rather than the statistically rare, high-intensity strikes that would occupy the upper reaches of the intensity distribution.

06 Incident Type Distribution

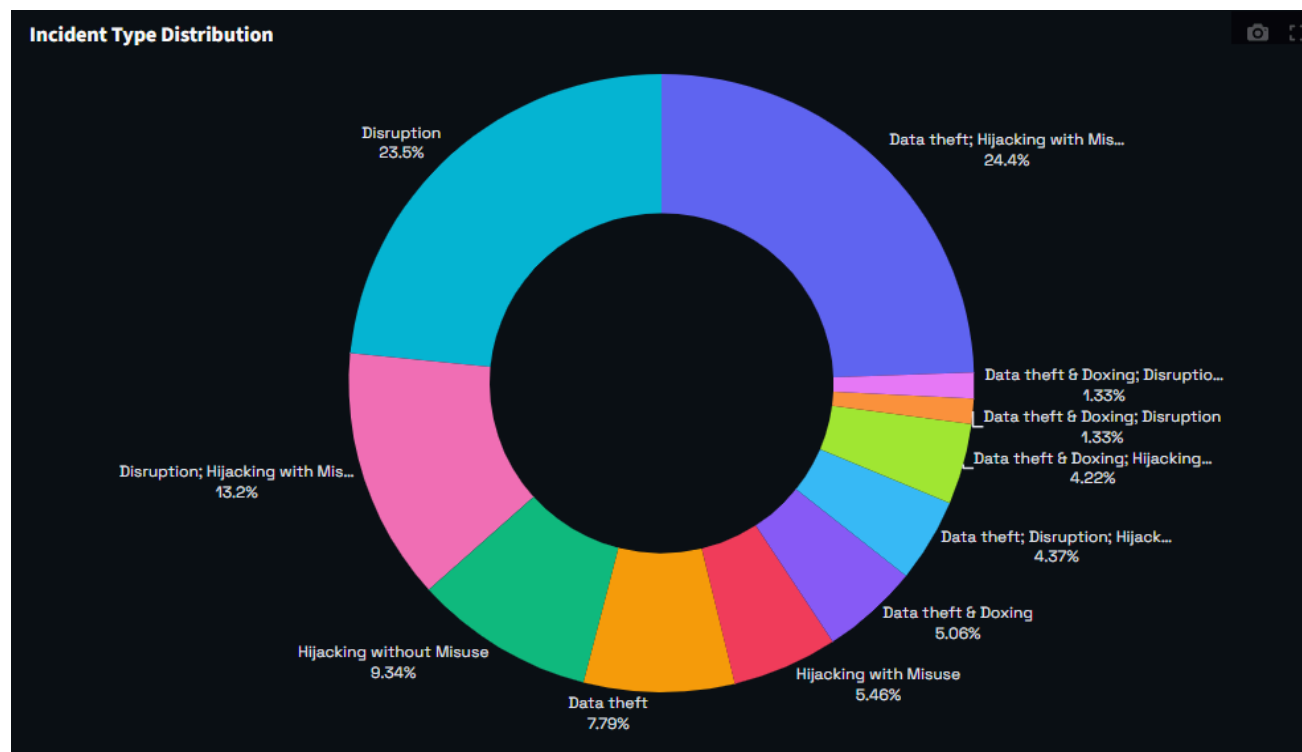


Figure 6. Incident Type distribution.

The visualization of "Incident Type Distribution" reveals a landscape where **Data theft; Hijacking with Misuse** (24.4%) and **Disruption** (23.5%) function as the dominant tactical choices, collectively comprising nearly 50% of all documented global cyber activity.

◆ KEY INSIGHT

The data reveals that modern cyber operations are frequently multi-faceted and complex. Instead of executing single-objective attacks like pure data theft (which sits lower at 7.79%), threat actors heavily favor compound attacks—most notably combining data exfiltration with the active misuse of compromised infrastructure. Furthermore, the high prominence of pure "Disruption" indicates that causing operational downtime, sabotage, or denial of service remains a primary and highly favored standalone objective for attackers.

07 Mitre Attack

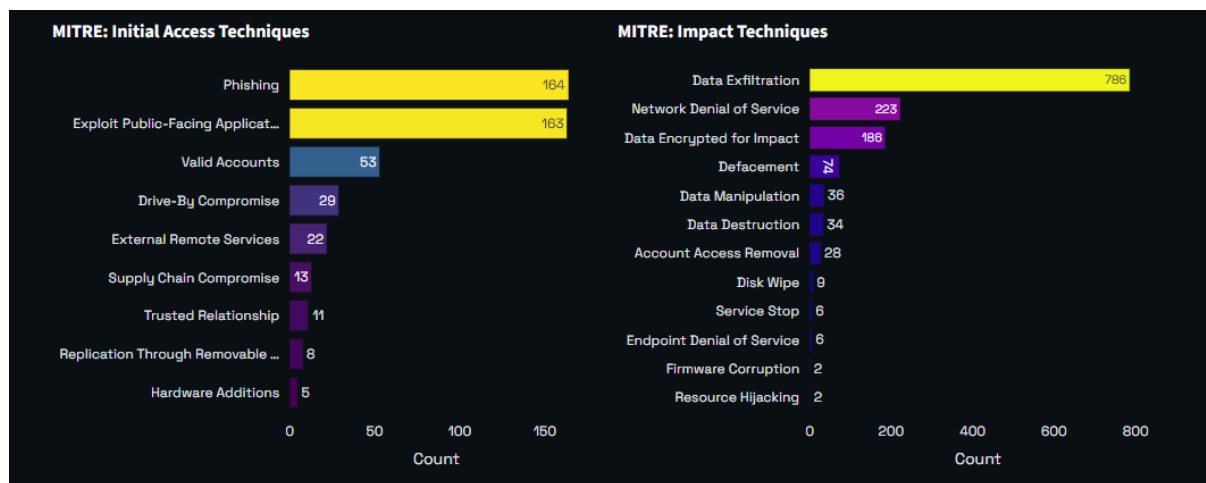


Figure 7. Mitre Attack.

The visualization of MITRE ATT&CK methodologies highlights a tactical landscape where **Phishing** (164 incidents) and the **Exploitation of Public-Facing Applications** (163 incidents) serve as the primary entry vectors, appearing with nearly identical frequency. Regarding operational impact, **Data Exfiltration** emerges as the dominant consequence with 786 documented cases, a volume that more than triples the 223 instances of Network Denial of Service.

◆ KEY INSIGHT

The analysis reveals a tactical landscape defined by a dual-vector approach, where **phishing** and the **exploitation of public-facing applications** are utilized with equal frequency to penetrate networks. Subsequent to entry, the overwhelming prevalence of **data exfiltration** indicates that the theft of information—whether for geopolitical espionage, intellectual property acquisition, or complex extortion—serves as the primary strategic objective for most adversaries, a volume that significantly exceeds standalone disruptive efforts.

o8 Countries issuing Legal Response

Countries Issuing Legal Responses

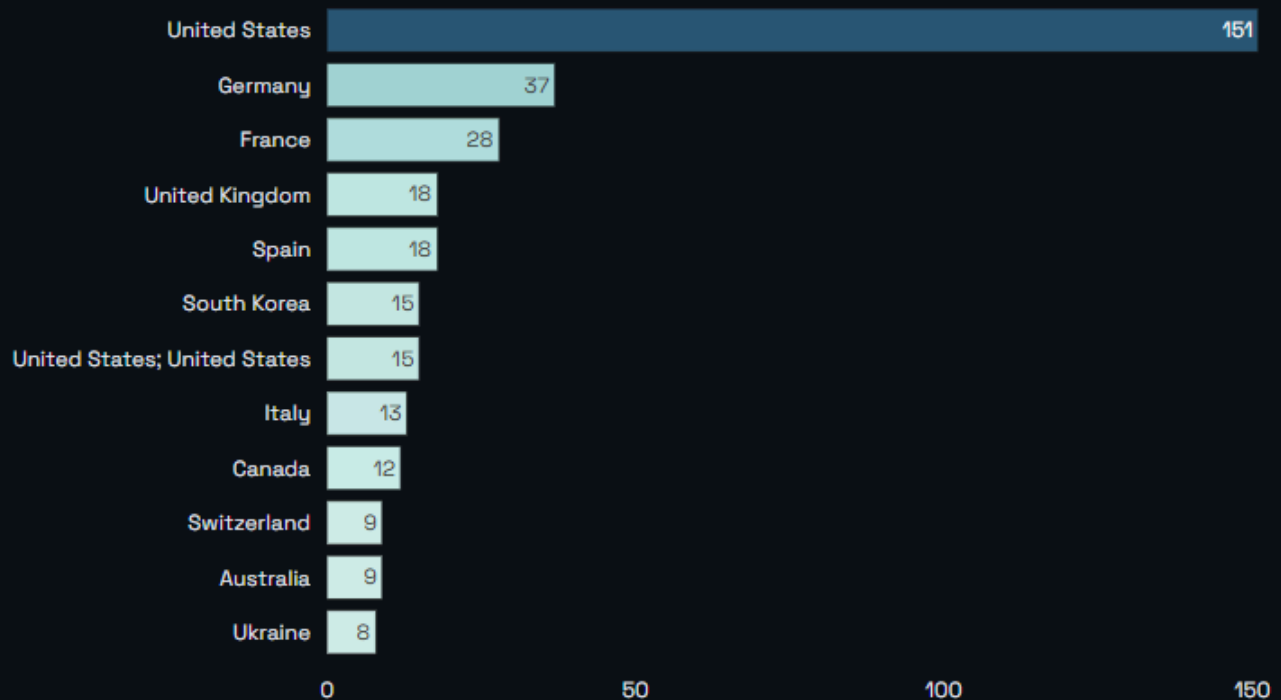


Figure 8. Country issuing Legal Response.

The United States maintains an unparalleled lead in pursuing legal recourse, recording 151 documented responses—a volume that exceeds the activity of Germany, the second-ranked nation with 37 instances, by more than fourfold.

◆ KEY INSIGHT

Despite a notable cluster of activity among European partners like Germany, France, and the UK, the extreme gap between the US and all other nations highlights a unique strategic preference. The United States appears to leverage formal indictments and domestic judicial processes as a core instrument of cyber diplomacy and discouragement at a rate that significantly outpaces that of its international counterparts.

09 Annual Incidents

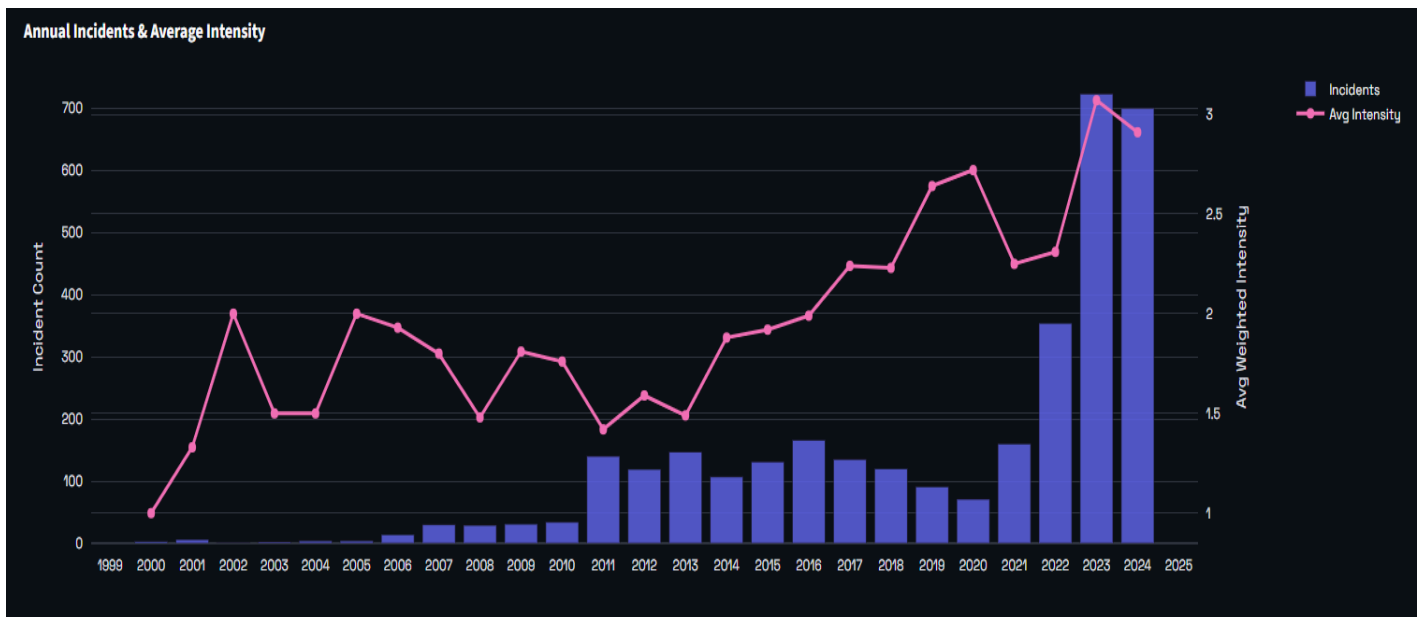


Figure 9. Annual Incidents.

The visualization of annual trends reveals a historical baseline of minimal activity before 2010, transitioning into a decade of steady, incremental growth. This landscape shifts dramatically starting in 2021, characterized by a staggering escalation in offensive volume—culminating in nearly 700 recorded cases by 2022/2023—while the **Weighted Intensity** simultaneously climbs to an unprecedented peak of approximately 3.0 during this same timeframe.

KEY TAKEAWAYS

- **Escalating Threat Landscape:** Cyber incident volume has surged dramatically since 2021, with weighted intensity peaking simultaneously. Historical data before 2015 should be interpreted as under-reported rather than reflecting a genuinely quiet period.
- **Tactical Shift Toward Complexity:** Adversaries increasingly favor compound attacks. Data theft combined with hijacking/misuse, and standalone disruptive attacks (sabotage/DoS), dominate the incident landscape, accounting for nearly 50% of all activity.
- **Reliance on Known Vulnerabilities:** The "zero-day" threat is overstated in frequency; over 97% of recorded incidents utilize known vulnerabilities, misconfigurations, or psychological manipulation (phishing) rather than rare, costly-to-procure novel exploits.
- **Low-Severity Dominance:** The threat environment is defined primarily by persistent, moderate-impact activities—such as routine intelligence gathering—rather than high-intensity, catastrophic strikes.
- **Gap in Attribution and Reporting:** A significant portion of the data (1,433 incidents) lacks clear functional or intelligence impact classification, indicating a substantial reporting gap. Furthermore, attribution relies heavily on media and industry reporting rather than formal government findings, requiring a cautious interpretation of the data.
- **Legal Response Imbalance:** The United States leverages formal judicial processes and indictments at a rate over four times higher than its closest international counterparts, suggesting a unique strategic preference for using legal recourse as an instrument of cyber diplomacy.